SNEH DESAI

DATA SCIENCE | MACHINE LEARNING | DATA ANALYTICS

Greenville, South Carolina | +1 864-356-8926 | sneh14022002@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Data science and analytics professional and M.S. Informatics student with a B.Tech. in Computer Science & Engineering (AI/ML). Builds end-to-end machine learning workflows in Python, from exploratory analysis and feature engineering through model evaluation, calibration, explainability, and Streamlit deployment. Experienced in SQL, PostgreSQL, Power BI dashboards, and communicating data-driven insights.

TECHNICAL SKILLS

Programming & Data: Python, SQL, PostgreSQL, pandas, NumPy, Microsoft Excel
Machine Learning: scikit-learn, Gradient Boosting, model comparison, hyperparameter tuning, probability calibration, threshold optimization, class-imbalance handling, SHAP
Analytics & Visualization: Exploratory Data Analysis, feature engineering, Matplotlib, Power BI, SAP Analytics Cloud, Streamlit, pytest, Jupyter Notebook, Git/GitHub

PROJECTS

F1 Race Strategy & Performance Intelligence Platform | Python, PostgreSQL, SQL, scikit-learn, Power BI

- Developed an end-to-end Formula 1 analytics platform analyzing historical race results, qualifying performance, championship standings, and finishing-position trends.
  - Designed a PostgreSQL data model and reusable SQL views for race, driver, constructor, qualifying, and championship analytics; built Power BI dashboards for championship analysis, qualifying vs. race pace, and race-strategy-related insights.
  - Compared ML models and selected a Gradient Boosting Regressor; recorded evaluation on the 2017 test season: MAE 3.61 positions, RMSE 4.59, R^2 0.367. Results are a historical demonstration, not a live race forecast.
  GitHub: github.com/Sneh14202/f1-race-intelligence

Customer Churn Prediction & Retention Intelligence Platform | Python, scikit-learn, SHAP, Streamlit

- Built an end-to-end churn prediction workflow on 7,043 telecom customer records, including EDA, feature engineering, model comparison, tuning, probability calibration, and threshold optimization.
  - Evaluated a calibrated Gradient Boosting model with 0.846 ROC-AUC, 0.674 PR-AUC, 81.3% recall, and 0.626 F1; segmented 28.3% of customers as high/critical risk.
  - Added SHAP-based explanations and business-value thresholding, with an interactive Streamlit risk-scoring app.
  GitHub: github.com/Sneh14202/customer-churn-intelligence

Fraud Detection & Transaction Risk Intelligence Platform | Python, scikit-learn, SHAP, Streamlit

- Built a fraud detection pipeline on 284,807 credit card transactions, addressing severe class imbalance through model comparison, tuning, calibration, and threshold optimization.
  - Engineered transaction/time features and reusable risk-scoring and inference modules backed by automated pytest tests; added SHAP explanations, five-level risk segmentation, and investigation prioritization.
  GitHub: github.com/Sneh14202/fraud-detection-intelligence

Power BI Dashboards — Game Sales, Adidas Sales & Swiggy Orders | Power BI

- Created interactive dashboards exploring video-game sales by genre/year, Adidas sales by product/season, and Swiggy orders by city/cuisine.
  GitHub: github.com/Sneh14202/Power-BI-Dashboards

EXPERIENCE

Data Analyst Intern, Institutional Planning and Research — University of South Carolina | Mar 2026–Present

- Analyze and organize institutional data; prepare reports and visual summaries to communicate trends supporting planning and research.

Graduate Assistant, Student Success Center — University of South Carolina | Jan 2026–Present

- Support student-focused programs, coordination, administrative tasks, and organized records in collaboration with campus stakeholders.

Math Tutor, Math Lab — University of South Carolina | Aug 2026–Present

- Explain mathematics concepts and guide university students through structured problem-solving.

Summer Intern, Data Analytics — Edunet Foundation | Jul 2023–Aug 2023

- Worked on practical analytics projects using Power BI and SAP Analytics Cloud, with exposure to machine learning, big data, and cloud computing.

EDUCATION & CERTIFICATIONS

M.S. in Informatics — University of South Carolina Upstate | 2025–Present
B.E./B.Tech. in Computer Science & Engineering (AI/ML) — RNG Patel Institute of Technology | 2020–2024
Certification: IBM SkillsBuild — Data Analytics (2-week virtual internship) | **Languages:** English, Hindi, Gujarati